

Name: _____

Euler's formula and cross sections.

1. A polyhedron has 9 faces and 9 vertices. How many edges?
2. A tetrahedron has 4 faces and 4 vertices. Check Euler's formula.
3. What cross section do you get slicing a cylinder parallel to its base? Perpendicular through the axis?

Volumes of prisms.

4. A rectangular prism is 7 by 4 by 3. Find its volume.
5. A prism's base is a triangle with base 9 and height 4; the prism is 11 units tall. Find V.

Volumes of cylinders. Exact and nearest tenth.

6. Radius 6, height 10.
7. Radius 4.5, height 8.

Working backward.

8. A prism with a 6-by-5 base has volume 180. Find its height.
9. A cylinder with radius 5 has volume 300π . Find its height.
10. A concrete step design stacks two blocks: $4\text{ m} \times 1\text{ m} \times 0.5\text{ m}$ and $4\text{ m} \times 0.5\text{ m} \times 0.5\text{ m}$. How much concrete is needed?